

Aarbakke AS
Jaervegen 142,
Port 2,
4344 ByrnesPostboks 108,
Bryne,
4349
Norway

Cert No:	EUR-472273	Cast No	329277
Sales Order No:	SO-EUR-720712-1	Cust Ref	03624
Cust Order No:	105320 - 1	Pieces	1
Size:	101.6MM DIAMETER X 1000MM LONG	Cert Date	03-Sep-24
Part No:			

Specification	ALLOY 625 TO M40409 REV E	Grain Size	Min: 9	Max: 9
Melt Practice	VIM/ESR	Hot Work Ratio	14.33	

Cast Analysis wt %

C	Si	Mn	P	S	Cr	Mo	Ni	Al	Ti	[Nb+Ta]	Co	Fe
0.0220	0.1800	0.0800	0.0060	0.0010	22.2700	9.1200	59.8000	0.2300	0.3400	3.4000	0.0200	4.4400

Mechanical Properties Test Piece (QTC) Size 4.000inch DIA

Test Number 513570 Test Piece Nature Integral

Tensile - MPa

	Location	Direction	PS 0.2%	UTS	Elongation %	R of A %
Req't	1/4t	Longitudinal	414 Min	693 Min	30 Min (4d)	40 Min
Result	1/4t	Longitudinal	450	900	53.8	58

Heat Treatment

Treatment	Temp	Soak	Transfer Time	Coolant	Charge	Start	Peak	End	Load	Thermocouple Practice
Anneal	920 °C	0 h 30 m		Water Quench		29 °C	30 °C	29 °C		Heat Sink

	Req'd	Min	/	Max	Achieved	Min	/	Max	
Material Hardness (Test EUR)		240		320		246		246	Hbw
QTC Hardness		240		320		266		268	Hbw

Additional Comments

CERTS TO BSEN 10204:2004 3.1
NACE MR-01-75
NI=BAL
NO WELD REPAIRS PERFORMED ON THIS MATERIAL
PRIMARY MILL - HEAT TREATED AND TESTED: VDM METALS
ROLLED: BOHLER
COUNTRY OF ORIGIN: AUSTRIA
BATCH NO 720712-1
SERIAL NO 720712-1-1
Q02324 REV M

All furnace Calibration conforms to API6A 21st Edition ANNEX M. Hardness testing: HBW = ASTM E10
Temperature of the water quench bath did not exceed 49 deg c at any time during the quench cycle.

Inspection certificate 265394/0
DIN EN 10204/01.05 3.1
Order No. 625056 Purchase Order No.
EUR-61385 (00251)

Page: 1 / 2
printed: 31. JUL 2024

Howco Group plc.

1-3 Woodside Crescent
GB - G3 7UL Glasgow



Trademark
VDM® Alloy 625
NICROFER 6020 HMO
2.4856

Product
Bar, round, hot rolled, annealed, machined

Specification

Howco HG 625 Rev. 7 (bars < 120 mm are not homogenised)
with exceptions as described in the template from 29.09.2020
ASTM B 446-19
ASTM B 564-22 applicable for forgings
ASME BPVC.II.B-2023 SB-446
ASME BPVC.II.B-2021 SB 564 applicable for forgings
ANSI/NACE MR0175-2021/ISO 15156-3:2020
AMS 5666 K (> 254 mm AMS 5666 K EXC)

Material

UNS N06625
UNS N06625
UNS N06625
UNS N06625
UNS N06625

HOWCO

CERTIFIED TRUE COPY

- 3 SEP 2024.

PAGE 2 OF 11 CERT NO 62293
SIGNED 3

Item	Pcs	Weight [kg]	Dimension [mm]	Heat	Lot	Delivery No.	
16	2	714	101,60 Ø x 3000-8000	329277	105548658	8682140015	4.00" dia.

Analysis (weight %)

Heat	country of melt
329277	Germany

LE=combustion analysis analysed corresponding to ASTM E1019-18

RF=X-ray fluorescence spectroscopy analysed corresponding to ASTM E572-21, ASTM E1085-22, ASTM E1621-22, ASTM E2465-23

Nb+Ta =

Howco HG 625 Rev. 7 (bars < 120 mm are not homogenised); ASTM B 446-19; ASTM B 564-22 applicable for forgings; ASME BPVC.II.B-2023 SB-446; ASME BPVC.II.B-2021 SB 564 applicable for forgings; AMS 5666 K (> 254 mm AMS 5666 K EXC)

Heat	Melting Pr.		C	S	Cr	Ni	Mn	Si	Mo	Ti	Nb	Cu	Fe
329277	VIM/ESR	top	0,022	<0,001	22,27	R59,80	0,08	0,18	9,12	0,34	3,39	0,02	4,44
		Melth	LE	LE	RF		RF	RF	RF	RF	RF	RF	RF

Heat	Melting Pr.		P	Al	Co	Ta	Nb+Ta
329277	VIM/ESR	top	0,006	0,23	0,02	<0,01	3,40
		Melth	RF	RF	RF	RF	

Subcontractor	Street	location	practice
voestalpine Böhler Edelstahl GmbH	Mariazellerstraße 25	8605 Kapfenberg / Austria	hot rolling

Sample Condition annealed		Tensile Test 1 ASTM E8/E8M - 22 / longitudinal specimen Temp Yield0.2% Yield1.0% Tensile Elongallon [°C] [MPa] [MPa] [MPa] A [%] A [%] RoA [%]										Grain size 1 ASTM E112 - 13 (Reappr. 2021)				
Lot	Sample No.											Test Location				
105548658	101T002-A07	1	RT	450			900	4D	53,8			58	1	No 9	1/4 thickness	
Sample Condition annealed		Corrosion Test 1 ASTM G 28-22 Melhod A - delivery condillon										Microstructure/Precipitates 1 Howco HG 625 Rev. 7 (bars < 120 mm are not homogenised)				
Lot	Sample No.											Test Location				
105548658	101T002-A07	1 without objection : 1,19 mm/a IK-attack = 145,0 µm										1		Complying with delivery specification.		1/4 thickness

We hereby certify, barring exceptions, reservations, or exemptions listed on this statement of conformity, that the listed supplies comply with the contract requirements and that, after completion of all contractually owed testing and verification, they completely satisfy all agreed requirements, standards and regulations.

Enclosures: 3

E-Mail christopher.schleif@vdm-metals.com

Christopher Schleif, authorized inspection representative

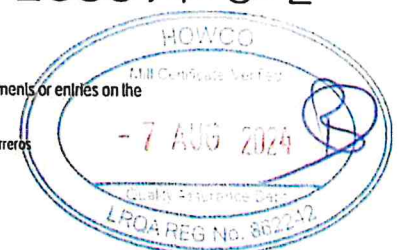
Date of release: 31.07.2024

This certified test report shall not be reproduced except in full, without the written approval of VDM Metals. The recording of false, fictitious or fraudulent statements or entries on the certificate may be punished as a felony under federal law.

VDM Metals GmbH
Division Bar and Forgings
Plettenberger Straße 2, D-58791 Werdohl
Postfach 18 20, D-58778 Werdohl
Telephone: +49 2392 550
Internet: www.vdm-metals.com

Chairman of the Supervisory Board: Bernardo Velázquez Herreros
Management Board: Dr. Nils Müller (Chairman),
Holger Arnold, Tobias Coerschte, Frank Morris
Commercial register: Iserlohn local court, #5327
Company domicile: Werdohl

* 265394- O- E *



Inspection certificate 265394/0
DIN EN 10204/01.05 3.1
Order No. 625056 Purchase Order No. EUR-61385 (00251)

Page: 2 / 2
printed: 31. JUL 2024

Values final control

cast	Part(s) ID
329277	101

Item	Lot	
16	105548658	Ultrasonic Examination: ok, details see enclosure

Heat Treatment	Lot	reduction ratio	Lot
920°C/00:30h/water-cooling[start 29°C - end 29°C - max. 30°C]	105548658	14,33:1	105548658

Spectrographic analytical mixup test/PMI check: without objection
Dimensional and visual inspection performed: without objection

No mercury contamination; no Radium, Luminous or Alpha source materials were used in the manufacture or testing of the items furnished.

No weld repairs performed.

Material is in accordance to the requirements of the requested specification.

For dimensions greater 20mmØ we use samples for tensile testing, that are standard ASTM specimens with 12.5 mm diameter, taken from a prolongation of the bars.

For dimensions greater 16mmØ, we use samples for notched-bar impact testing, which are standard ASTM specimens with 10 mm x 10 mm square, taken from a prolongation of the bars.

SAMPLE LOCATION:

all samples for mechanical tests (without hardness) were taken from ¼ T for >Ø 50 mm;
all samples for mechanical tests (without hardness) were taken from ½ T for <=Ø 50 mm.

The pyrometry of heat treatment is in accordance with AMS 2750 latest revision.

Grade 1

MACROETCHING: conducted as per ASTM A 604,
sample taken from top and bottom of ingot after hot forming.

Rating: class1 - class 4 better or equal - severity B

Melting location: VDM Metals, Formerstraße 17, D-59425 Unna / Germany

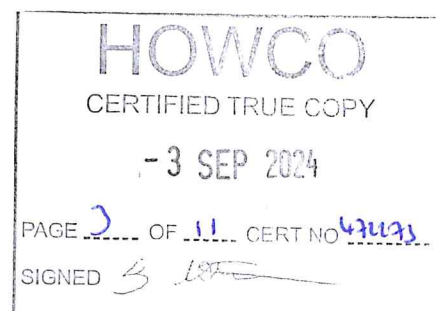
VDM Metals International GmbH has complied with all producer requirements of AS6279 and has instructed and received confirmation by its outsourced toll manufacturer VDM Metals GmbH accordingly.

6.2.3.2.: Grain boundary carbides, continuous boundary carbides or other secondary phases noted as present along grain boundaries but the G 28 A test result confirms their presence is not detrimental to quality. The material is free from laves phase

12.1.: Access to subcontractors is only allowed in exceptional circumstances

Heat treatment furnace TUS performed annually, associated instrumentation calibrated quarterly, both in accordance with API 6A Annex M

Certified quality management system for material manufacturers according to pressure equipment directive 2014/68/EU, Annex I, point 4.3 - TÜV NORD Systems GmbH & Co. KG, Certification Body for Pressure Equipment, Notified Body, Code no. 0045



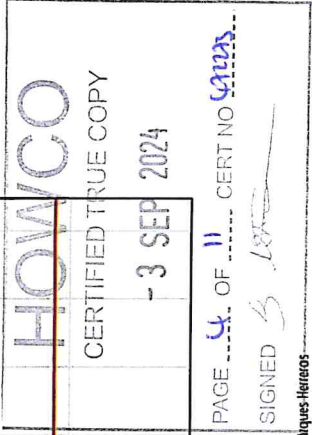
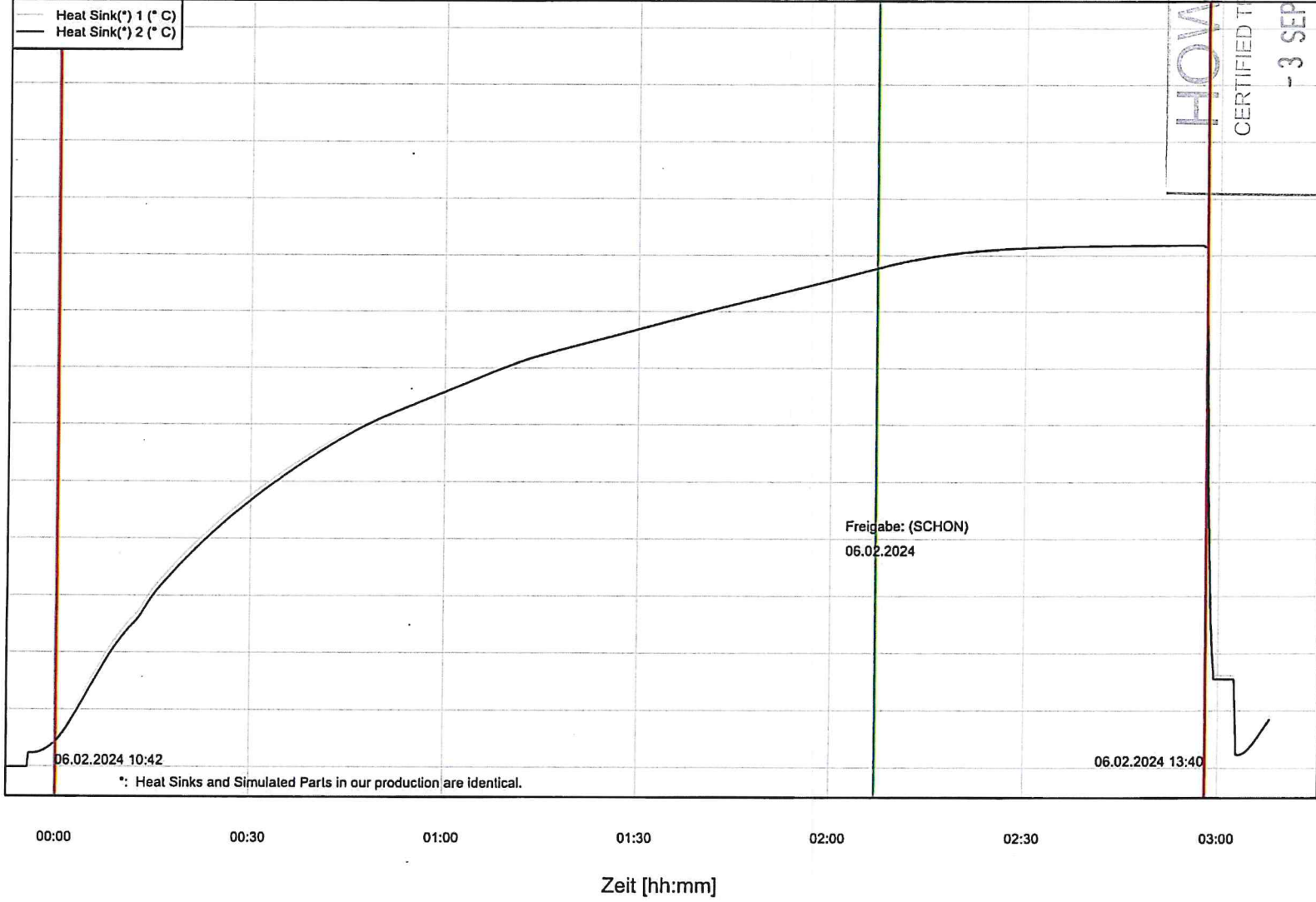
Enclosure for Inspection certificate
DIN EN 10204/01.05

265394/0
3.1

Enclosure:
Page:
printed:

2
1/1
30.JUL.2024

1300 °C
(2372 °F)
1200 °C
(2192 °F)
1100 °C
(2012 °F)
1000 °C
(1832 °F)
900 °C
(1652 °F)
800 °C
(1472 °F)
700 °C
(1292 °F)
600 °C
(1112 °F)
500 °C
(932 °F)
400 °C
(752 °F)
300 °C
(572 °F)
200 °C
(392 °F)
100 °C
(212 °F)
0 °C
(32 °F)



Chairman of the Supervisory Board: Bernardo Velázquez-Hereros
Management Board: Dr. Heiko Müller (Chairman)
Höger Arnold, Tobias Coerschte, Frank Moritz
Commercial register: Berlin local court #5327
Company domicile: Werdohl

VDM Metals GmbH
Division Bar and Forgings
Pellenberger Straße 2, D-58791 Werdohl
Postbox 1620, D-58778 Werdohl
Telephone: 02392 55-0, Fax: 02392 55-2217
Internet: www.vdm-metals.com

Enclosure for Inspection certificate 265394/0
DIN EN 10204/01.05 3.1
Order No. Delivery No. Purchase Order No.
625056
EUR-61385 (00251)

Enclosure: 1
Page: 1 / 2
printed: 31. JUL 2024

Hardness Test						
annealed		ASTM E10 - 18				
Lot	Sample No.	Location	Method	Values		
105548658	101T003 -A07	surface	HBW 2.5/187.5 /A	266	266	268

Notched Bar Impact Testing										
annealed		ASTM E23 - 23a Charpy-V sample / longitudinal (L-R) / KV ₀ 450/10								
Lot	Sample No.	Temp. [°C]	KV [J]		aK [J/cm²]	Lateral Expansion [mm]			Shear Fracture [%]	
105548658	101T002 -A07	-60	123,4	123	124,1	1,33	1,34	1,34	100	100

HOWCO
CERTIFIED TRUE COPY
- 3 SEP 2024
PAGE 5 OF 11 CERT NO 42273
SIGNED 

Enclosure for Inspection certificate
DIN EN 10204/01.05

265394/0
3.1

Order No.
625056

Delivery No.

Purchase Order No.

EUR-61385 (00251)

Enclosure: 1
Page: 2 / 2
printed: 31. JUL 2024

Ultrasonic Examination Report

Specification

ASTM A388/A388M - 19
US-test acc. to API 6A 21st Edition PSL 3 & PSL 4 (10.4.2.12.2 & 10.4.2.12.3)
API 6A PSL3 & PSL4 100%

Inspection Technique	straight beam in radial direction
Surface Condition	machined
Inspection Scope	API 6A PSL3 & PSL4 100%
Test Apparatus	Karl Deutsch Echograph 1095/ 11134
UT Device: Calibration date / due date	03/24 - 09/24
Couplant	water
Type of Probe / Frequency / Dim.	PL2 205735 / 2,00 MHz / 24,0 mm
Calibration Block/sound path	TK 26: 39,6 / 70,2 / 102,4 [mm]
Recording Level	2,26 mm
Acceptance Level	3,20 mm
Scanning Speed	150,00 mm/Sek.

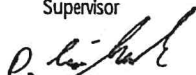
Cast - Product-ID	Reportable Indications		
329277 - 101	count	Inspector	level
	0	Schon	Level II

Date of Inspection

17.07.2024

The material is in compliance with the test specification

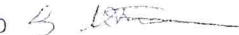
Supervisor


Hirzbruch, Level II

HOWCO
CERTIFIED TRUE COPY

- 3 SEP 2024

PAGE 6 OF 11 CERT NO 47273

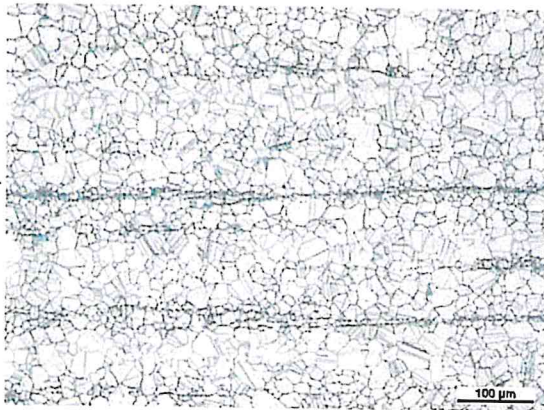
SIGNED 

Microstructure Examination

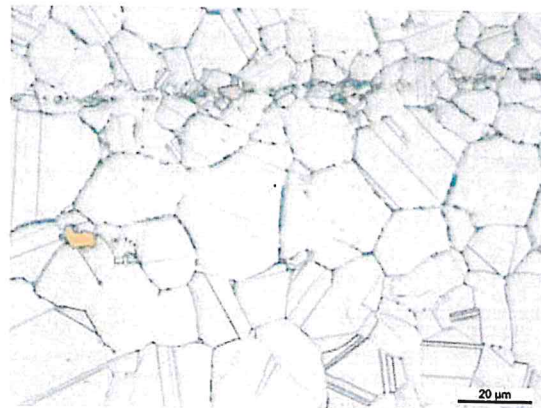
Metallography Job No.: 202050
Lot No.: 105548658
Heat: 329277

Inspection Certificate No.: 265394

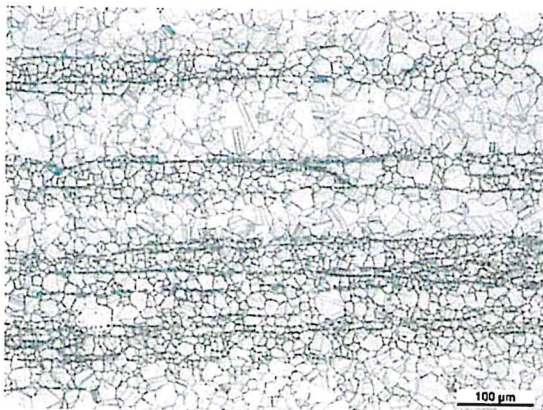
Enclosure: 3
Page: 1 of 1



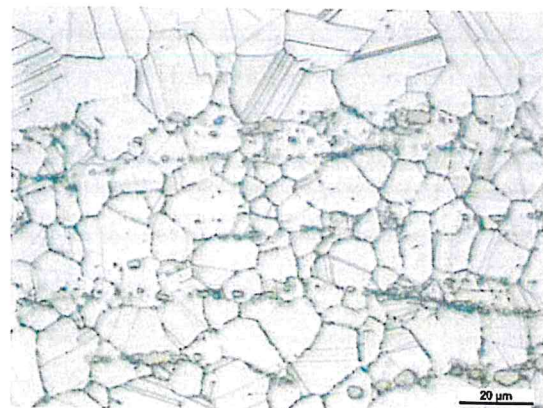
100:1 VDM® Alloy 625 /Nicrofer 6020 hMo
SMT
HCl - HNO₃ 6:1 24.07.2024 11:02:48
101T002-A07 typical 1/4 thickness



500:1 VDM® Alloy 625 /Nicrofer 6020 hMo
SMT
HCl - HNO₃ 6:1 24.07.2024 11:02:49
101T002-A07 typical 1/4 thickness



100:1 VDM® Alloy 625 /Nicrofer 6020 hMo
SMT
HCl - HNO₃ 6:1 24.07.2024 11:02:51
101T002-A07 worst 1/4 thickness



500:1 VDM® Alloy 625 /Nicrofer 6020 hMo
SMT
HCl - HNO₃ 6:1 24.07.2024 11:02:52
101T002-A07 worst 1/4 thickness

Address: VDM Metals GmbH, Kleffstrasse 23, D-58762 Altena, P.O. Box 12 51, D-58742 Altena
Telephone: +49 2392 55-0 Fax: +49 2392 55-2217 Internet: www.vdm-metals.com
Chairman of the Supervisory Board: Bernardo Velázquez Herreros
Management Board: Dr. Niclas Müller, Chairman, Tobias Coerschulte, Frank Morris, Holger Arnold
Company domicile: Werdohl Commercial register: Iserlohn local court, HRB 5327

HOWCO

CERTIFIED TRUE COPY

- 3 SEP 2024

PAGE 7 OF 11 CERT NO. 4700

SIGNED 3



Brinell Hardness Test Report

CUSTOMER: Aarbakke AS
PRODUCT CODE: 625 4 ANN

ORDER NUMBER: 720712-1
STEP: 752768
HARDNESS RANGE (HBW): 240.0 - 320.0

EQUIPMENT TYPE / SERIAL NO.

King	Avery	Microscope	Brintronic	Another (Equotip)	Rockwell HRC	Foundrax Static
☒ UC-71	☐ 6403-4 R92033	☐	☐ 0119070	☐	☐ F68332/139	☐
☐ UC-72	☐ 6403_16897	☐	☒ 0510014	☐	☐	☐
☐	☐ E50152/15 (6503-1)	☐	☐ 0907068	☐	☐	☐
☐	☐ HMV1 (6403-3)	☐	☐ 1023172	☐	☐	☐
☐	☐ R335 (HMM1)	☐	☐ 1112037	☐	☐	☐

Nature of Test Piece: Integral

CAST NUMBERS

Cast: 329277

Min:

Max:

TAGS

1341355 QTY: 1

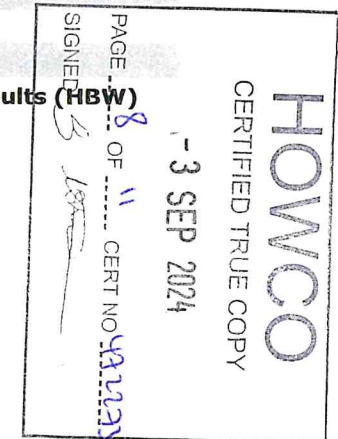
Material Hardness Check Results (see drawing to reference the numbers)

Result (HBW) / (HBW/P) - * Delete as applicable

									T/P Results (HBW)
1	246	11	21	31	41	51	61	71	TP1
2		12	22	32	42	52	62	72	TP2
3		13	23	33	43	53	63	73	TP3
4		14	24	34	44	54	64	74	TP4
5		15	25	35	45	55	65	75	TP5
6		16	26	36	46	56	66	76	TP6
7		17	27	37	47	57	67	77	TP7
8		18	28	38	48	58	68	78	TP8
9		19	29	39	49	59	69	79	TP9
10		20	30	40	50	60	70	80	TP10

Avery Brinell machines calibrated by Denison Mayes Group. King Brinell machines calibrated and test blocks supplied by Foundrax Eng Products Ltd. Denison Mayes Group, accredited for calibration by UKAS to ISO/IEC 17025, approval/schedule number 0066. Foundrax Engineering Products Ltd, accredited for calibration by UKAS to ISO/IEC 17025, approval/schedule number 0231. Material tested to specific customer hardness specification and to general requirements of ASTM E10 and BS EN ISO 6506-1/2 and NACE MR0175. Latest revisions are applicable. Hardness load/penetration depth - HBW 10 diameter (mm)/3000 kgf test force per ASTM E10. Hardness test performed at ambient temperature (18° to 35° Celsius). Hardness checks performed using a portable hardness test machine have been performed in accordance with ASTM E110.

Operator Signature	Test Block Standard (HBW)	Day of Use Value (HBW)	Test Block/Serial	Pass/Fail	Date
	313	310.5	28485/30AW	P	219



ULTRASONIC EXAMINATION

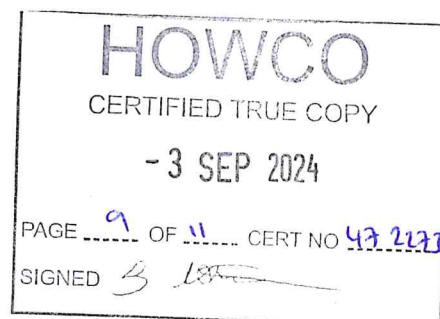
Customer:	Aarbakke AS	Order Number:	720712 - 1
Customers Order No:	105320-1	Report Number:	720712 - 1 /U
Examination Procedure:	Q02324 REV M	Tag Number:	1341355
Acceptance Procedure:	Q02324 PSL3 REV M	Cast Number:	329277
Material Grade	625	Quantity Tested:	1
Sales Order Description:	101.6MM DIAMETER X 1000MM LONG	Report Date:	Saturday, August 31, 2024
Part No:		Test Date:	Saturday, August 31, 2024 11:18
Serial No:	720712-1-1		
Batch No:	720712 - 1		

Method	Probe, Frequency & Type		Serial	Test Range	Method of Sensitivity & Test Block Serial Number	Test Gain
1	GE-B2S 24MM DIA 0 DEG	2MHz SINGLE COMPRESSION	77486	0-1200MM	1/4" FBH DAC A718-SET 3-16	50+6dB
2	GE-B2S 24MM DIA 0 DEG	2MHz SINGLE COMPRESSION	77486	0-150MM	1/8" FBH DAC A718-SET 2-8	60+6dB
3	GE-WB45-2 20X22MM 45 DEG	2MHz SINGLE SHEARWAVE	55308	0-400MM	3% NOTCH A718-101.5WT-3T	50+6dB

Comments:

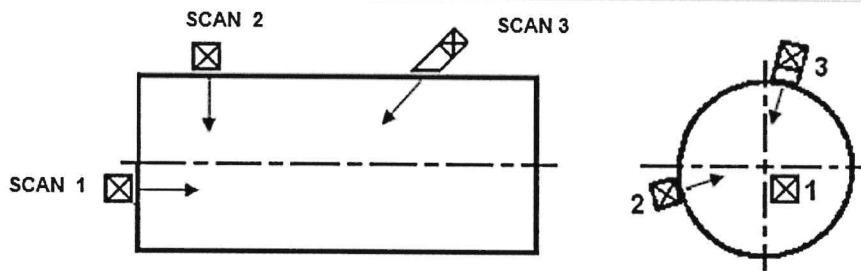
Visual Examination Performed & Acceptable to : Q02500 REV V & Q00499 REV D	
Examination carried out in accordance with the following approved scan plan: SPC10094224 REV D	
Coverage: 100% VOLUME + END - END	System: MANUAL SCANNING
Surface Condition: PEELED	Equipment: SONATEST D20 S/N I013193
All Examinations Carried Out At Ambient Room Temperature	Calibration Date: 6 February, 2024
Examination performed from the outside surface	Ultrasonic Examination Performed from the OD & End Faces
Result: SATISFACTORY WITH NO RECORDABLE INDICATIONS	Damping & Reject settings: Not Used
Search Unit Cable Length & Type: 2 Metre Long Co-Axial Cable	
Couplant: SONAGEL D1	
Inspector: MARTYN LEE	Signature: Digitally Signed By: <i>M. Lee</i>
PCN Level II No 302289 Expires 20/08/2027	For and on behalf of Howco Group

Witness:



ULTRASONIC SCAN PLAN

Customer:	Aarbakke AS	Order Number:	720712 - 1
Customers Order No:	105320-1	Report Number:	720712 - 1 /SP
Examination Procedure:	Q02324 REV M	Tag Number:	1341355
Acceptance Procedure:	Q02324 PSL3 REV M	Cast Number:	329277
Material Grade	625	Quantity Tested:	1
Sales Order Description:	101.6MM DIAMETER X 1000MM LONG	Report Date:	Saturday, August 31, 2024
Part No:		Test Date:	Saturday, August 31, 2024 11:19
Serial No:	720712-1-1	Scan Plan:	SPC10094224 REV D
Batch No:	720712 - 1		

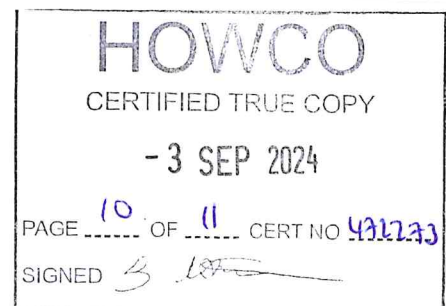


Scan	Coverage Of Scan	Probe, Frequency & Type	Serial	Sensitivity
1	100% COVERAGE FROM END FACES	GE-B2S 24MM DIA 0 DEG 2MHz SINGLE COMPRESSION	77486	1/4" FBH
2	360° X F/L OF SECTION	GE-B2S 24MM DIA 0 DEG 2MHz SINGLE COMPRESSION	77486	1/8" FBH
3	360° X F/L OF SECTION IN BOTH AXIAL DIRECTIONS	GE-WB45-2 20X22MM 45 DEG 2MHz SINGLE SHEARWAVE	55308	3% NOTCH

Comments:

Visual Examination Performed & Acceptable to :	Q02500 REV V & Q00499 REV D
Coverage: 100% VOLUME + END - END	System: MANUAL SCANNING
Surface Condition: PEELED	Equipment: SONATEST D20 S/N I013193
	Calibration Date: 6 February, 2024
All Examinations Carried Out At Ambient Room Temperature	Ultrasonic Examination Performed from the OD & End Faces
Examination performed from the outside surface	Damping & Reject settings: Not Used
Result: SATISFACTORY WITH NO RECORDABLE INDICATIONS	Search Unit Cable Length & Type: 2 Metre Long Co-Axial Cable
Inspector: MARTYN LEE	Couplant: SONAGEL D1
PCN Level II No 302289 Expires 20/08/2027	Signature: Digitally Signed By: <i>M. Lee</i>
	For and on behalf of Howco Group

Witness:



NDE Personnel Roster

Roster Qualified Level II & Level III Non-Destructive Testing Personnel.

Howco

Sheffield Department

Date: **6 October 2023**

Rev: **13**

Stamp No.	Name	MT	Expires	PT	Expires	UT – Manual Conventional	Expires	UT - PAUT	Expires
1	Alan Boyle PCN 212166	BINDT PCN LIII	29/06/2025	BINDT PCN LIII	14/06/2026	BINDT PCN LIII F PCN LIII W	14/09/2025 16/03/2026	BINDT PCN LII	04/04/2024
2	Michael Stansfield PCN 344145	BINDT PCN LII	21/09/2026	BINDT PCN LII	16/02/2025	BINDT PCN LII	16/11/2027	-	-
4	Stephen Smith PCN 309594	BINDT PCN LII	20/09/2026	BINDT PCN LII	13/09/2026	BINDT PCN LII	01/05/2025	-	-
5	Martyn Lee PCN 302289	BINDT PCN LII	TBC	BINDT PCN LII	-	BINDT PCN LII	20/08/2027	-	-
6	James Beattie PCN 317423	BINDT PCN LII	11/12/2027	BINDT PCN LII	18/12/2027	BINDT PCN LII	04/06/2028	-	-
7	Paul Jenkinson PCN 100727	BINDT PCN LII	22/01/2027	BINDT PCN LII	16/01/2027	BINDT PCN LII	14/06/2024	-	-

Compiled By: Alan Boyle • European NDE Manager • PCN Level III 212166

Howco

Carbrook Street Sheffield S9 2JN

Page 1

